Explain the TCP 3-way handshake and teardown, with a table of TCP flags (SYN, ACK, FIN, RST, PSH, URG)

1. What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented transport-layer protocol. Before two devices exchange application data, TCP establishes a connection using a process called the three-way handshake.

TCP provides:

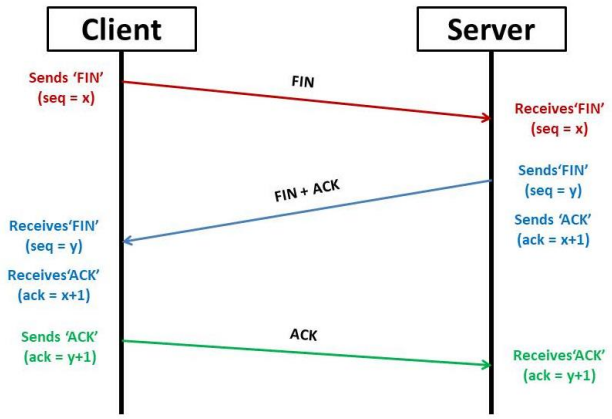
- Reliable data delivery.
- Ordered data transmission.
- Error detection and retransmission.
- Flow control.
- Connection management.

For example, when you open a website, your computer establishes a TCP connection with the web server (unless the connection uses a different transport protocol).

2. TCP three-way handshake

The three-way handshake establishes a TCP connection between a client and a server.

TCP 3-way handshake



Step-by-step explanation

Step 1: SYN (Synchronize)

- Client sends a SYN packet to the server.
- It requests a TCP connection.
- The client selects an initial sequence number (ISN), represented by x.

Example:

Source: 192.168.1.10:51500

Destination: 93.184.216.34:443
Flags: SYN
Sequence Number: 1000

Step 2: SYN-ACK (Synchronize-Acknowledgment)

- Server receives the SYN.
- Server responds with SYN + ACK.
- Server selects its own initial sequence number, represented by y.
- Server acknowledges the client's sequence number.

Example:

Source: 93.184.216.34:443
Destination: 192.168.1.10:51500
Flags: SYN, ACK
Sequence Number: 5000
Acknowledgment Number: 1001

Step 3: ACK (Acknowledgment)

- Client acknowledges the server's SYN.
- The TCP connection is established.
- Application data can now be transmitted.

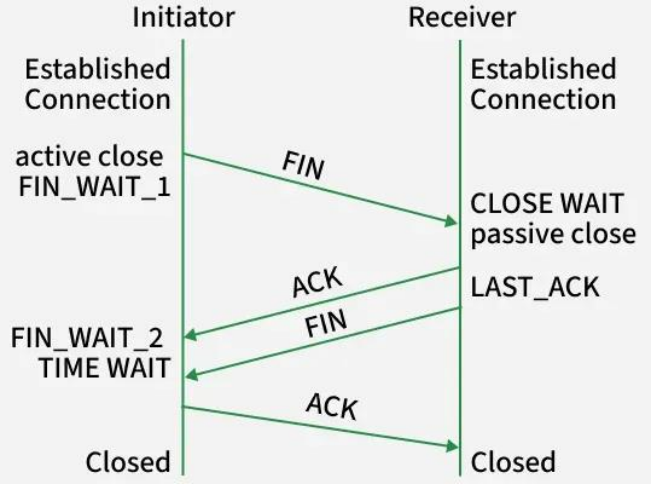
Example:

Source: 192.168.1.10:51500
Destination: 93.184.216.34:443
Flags: ACK
Sequence Number: 1001
Acknowledgment Number: 5001
Connection established

3. TCP connection teardown (4-way handshake)

TCP normally uses a four-segment exchange to close a connection gracefully.

TCP 4-way teardown



Step-by-step teardown

a. FIN: Client indicates that it has finished sending data.
b. ACK: Server acknowledges the client's FIN.
c. FIN: Server sends its own FIN when it has finished sending data.
d. ACK: Client acknowledges the server's FIN.

The client typically enters the TIME_WAIT state to ensure delayed packets do not interfere with a future connection and to allow retransmission of the final ACK if needed.

What is RST?

RST (Reset) terminates or rejects a TCP connection abruptly.
Example:

- A client attempts to connect to a closed port.
- The server responds with RST.
- The connection is rejected instead of being gracefully closed.

SOC relevance: A large number of RST packets may indicate port scanning, connection failures, firewall rejection, or application problems. Context is important.

4. TCP flags table

Flag	Full form	Purpose	SOC relevance
SYN	Synchronize	Starts a TCP connection	Identify connection attempts and SYN scans
ACK	Acknowledgment	Acknowledges received data or connection setup	Normal in established TCP traffic
FIN	Finish	Gracefully closes a connection	Identify normal or unusual connection termination
RST	Reset	Abruptly resets a connection	Identify rejected connections or abnormal termination
PSH	Push	Requests delivery of buffered data to the application promptly	Common in data-transfer traffic
URG	Urgent	Indicates urgent data using the urgent pointer	Unusual in many modern applications; investigate context

Important: TCP flags can be combined. For example:

- SYN + ACK: Server response during connection establishment.
- FIN + ACK: Common graceful connection closure.
- PSH + ACK: Common during application data transfer.
- RST + ACK: Connection reset with acknowledgment information.